

Assignment 16: Systems of Equations

DO NOT WRITE ON THIS PAPER

Part A

Consider the following system of equations.

$$\begin{aligned}x + 2y &= 5 \\ 3x - 2y &= -1\end{aligned}$$

Solve the system of equations using substitution or elimination. Show your work and the check step.

Part B

Consider the following systems of equations:

$$\begin{aligned}y - x &= 1 \\ 2y - 2x &= 2\end{aligned}$$

Use substitution or elimination to solve the system of equations. How many solutions does the system of equations have?

Part C

Consider the following systems of equations:

$$\begin{aligned}y &= x + 1 \\ y - x &= 2\end{aligned}$$

Use substitution or elimination to solve the system of equations. How many solutions does the system of equations have?

Question 1

Big Question: What does it mean to solve a system of equations?

Use the following words in your response: *equal, variables, intersection*

Then answer the following word problem:

Juan will rent a car for the weekend. He can choose one of two plans. The first plan has an initial fee of \$45 and costs an additional \$0.20 per mile driven. The second plan has no initial fee but costs \$0.70 per mile driven.

For what amount of driving do the two plans cost the same? What is the cost when the two plans cost the same?

Question 2 Work

Solve the following system of equations. Show your work and the check step.

$$\begin{array}{r} 3y + 9 = x \\ 3x + 2y = 5 \end{array}$$

Question 2 Answer

Write your answer
in your Class Notes

Question 3 Work

Melissa is a software saleswoman. Her monthly salary is \$2,200 plus an additional \$40 for every copy of History is Fun she sells. Let S represent her total salary this month (in dollars) and let N represent the number of copies of History is Fun she sells.

Write an equation relating S to N .

Then, use this equation to find how many copies of History is Fun she needs to sell to earn a total salary of \$6,520 this month.

Question 3 Answer

Write your answer
in your Class Notes

Additional Space